



# Differentiating between mild papilledema and pseudopapilledema on fundus photography

Marko Chi-Wei Tien<sup>a</sup>, Jim Xie<sup>b</sup>, Armin Handzic<sup>c</sup>, Laura Donaldson<sup>d</sup>, Edward Margolin<sup>b,e,\*</sup>

<sup>a</sup> University of British Columbia, Faculty of Medicine, Vancouver, British Columbia, Canada

<sup>b</sup> University of Toronto, Faculty of Medicine, Department of Ophthalmology and Vision Sciences, Toronto, Ontario, Canada

<sup>c</sup> University of Zurich, Faculty of Medicine, Zurich, Switzerland

<sup>d</sup> McMaster University, Department of Ophthalmology, Hamilton, Ontario, Canada

<sup>e</sup> University of Toronto, Faculty of Medicine, Department of Medicine, Division of Neurology, Toronto, Ontario, Canada

## ARTICLE INFO

### Keywords:

Papilledema

Fundus photography

Telemedicine

Idiopathic intracranial hypertension

Pseudopapilledema

## ABSTRACT

**Objective:** True papilledema can be difficult to distinguish from pseudopapilledema, especially if mild. Tele-ophthalmology allows fundus photos of suspected disc edema to be sent remotely to neuro-ophthalmologists for their expert interpretation. This study evaluated the ability of neuro-ophthalmologists to discriminate papilledema from pseudopapilledema in mild cases of optic disc swelling based on fundus photographs, with and without patient information.

**Methods:** Fundus photographs from 55 patients referred to two neuro-ophthalmology centers for suspicion of papilledema with mild optic disc swelling were reviewed. Patients with normal visual function and no evidence of secondary intracranial hypertension were included. Strict definitions were used to define papilledema and pseudopapilledema. Three neuro-ophthalmologists rated fundus photos as either papilledema or pseudopapilledema before and after the provision of patient age, sex, body mass index (BMI), and visual acuity.

**Results:** Twenty-nine patients fulfilled criteria for the papilledema group and 26 in the pseudopapilledema group. Fifty-one subjects were female, average age was 32.8 years, and average BMI was 30.9 kg/m<sup>2</sup>. When photos were rated in isolation, overall accuracy was 59.4 % and inter-rater reliability was poor ( $k = 0.052$ ). The true positive rate was 56.3 %, true negative rate 62.8 %, false positive rate 37.2 %, and false negative rate 43.7 %. Providing patient information improved overall accuracy to 64.2 % and inter-rater reliability to  $k = 0.37$ .

**Conclusions:** It is difficult even for neuro-ophthalmologists to correctly distinguish mild papilledema from pseudopapilledema based on fundus photographs alone. Providing basic patient information in telemedicine referrals can improve clinical appraisal and guide the invasiveness of investigations.

## 1. Introduction

In the rapidly evolving field of tele-ophthalmology, imaging reports and fundus photographs are essential in narrowing the differential diagnosis. Fundus photography allows for clearer visualization of ocular findings by non-specialists and can be distributed to ophthalmologists for expert interpretation [1,2]. Analyzing fundus photographs remotely has become commonplace for several ophthalmic subspecialties, including neuro-ophthalmology, wherein referrals to rule out disc edema are common [3,4]. Referring providers may include fundus photos to aid clinical decisions regarding work-up and the need for invasive procedures such as lumbar puncture (LP).

True disc edema, when accompanied by normal visual function in the typical demographic for idiopathic intracranial hypertension (IIH), is usually due to papilledema. Distinguishing papilledema and pseudopapilledema is crucial in ruling out diseases associated with intracranial hypertension that are vision- or life-threatening [5,6]. However, mild papilledema can be difficult to distinguish from pseudopapilledema based on ophthalmoscopy alone [7]. LP demonstrating an elevated opening pressure is required to diagnose IIH and can rule out inflammatory and neoplastic processes; however, it is an invasive procedure that carries several disadvantages, including 1) large body habitus making the procedure technically challenging, 2) patient positioning which may cause measurement error, 3) geographic access which may

\* Corresponding author at: University of Toronto, Dept. of Ophthalmology and Visual Sciences, Dept. of Medicine, Division of Neurology, 801 Eglinton Ave West, Suite 301, Toronto, Ontario M5N 1E3, USA.

E-mail address: [Edward.margolin@uhn.ca](mailto:Edward.margolin@uhn.ca) (E. Margolin).

<https://doi.org/10.1016/j.jns.2025.123482>

Received 12 January 2025; Received in revised form 28 February 2025; Accepted 27 March 2025

Available online 30 March 2025

0022-510X/© 2025 Published by Elsevier B.V.

be limited, and 4) patient refusal to pursue the LP. In systemically well patients who fit the typical demographic for IIH and demonstrate indirect neuroimaging signs of raised intracranial pressure (ICP), an informed decision with a neuro-ophthalmologist to treat based on a presumptive diagnosis of IIH may be reasonable [8–10]. Mild disc edema in otherwise healthy individuals without red flag symptoms or signs of intracranial hypertension should not immediately prompt invasive procedures [11–13].

Clinical suspicion for papilledema is difficult to gauge from fundus photographs alone if edema is mild and there is no evidence of pseudopapilledema such as superficial optic nerve drusen (ODD) or other congenital anomalies such as tilted optic discs or myelinated nerve fibers. While real-world clinical practice involves incorporating medical history, examination findings, and investigative results to arrive at a diagnosis, referral letters are often missing key clinical information such as visual acuity and body mass index (BMI) to aid in the neuro-ophthalmologist's recommendation on whether or not to pursue an LP. More importantly, in the evolving age of tele-ophthalmology, the ability to interpret fundus photos with limited clinical information becomes important to triage who should be seen and when. A previous study investigated clinicians' ability to detect true papilledema on fundus photographs alone, with neuro-ophthalmologists being the most accurate [14]. However, most cases of disc edema were moderate-severe and the impact of demographic and clinical information on grader accuracy was not investigated. Furthermore, other specialties may be hesitant to order LP in patients with disc edema who appear otherwise well and may request neuro-ophthalmic opinion to guide management. In this study, we aimed to evaluate whether neuro-ophthalmologists can differentiate between papilledema and pseudopapilledema secondary to ODD in mild cases of optic disc swelling captured on fundus photography, and whether the addition of clinical context improved their ability to perform this task.

## 2. Methods

### 2.1. Study design and participants

This cross-sectional study included patients who were referred to two tertiary university-affiliated neuro-ophthalmology practice for suspicion of papilledema with mild disc swelling (defined as average peripapillary retinal nerve fiber layer (RNFL) thickness of 150  $\mu\text{m}$  or less on peripapillary optical coherence tomography (OCT)) between January 2018 and September 2024. Inclusion criteria included normal visual acuity (VA) (logMAR 0.18 or better), full visual fields (VF, mean deviation [MD]  $\geq -5.00$  or better on automated perimetry), and no relevant confounding ophthalmic diseases nor suspected causes of pseudopapilledema other than ODD such as vitreopapillary traction and crowded optic discs. No patients had any known history of intracranial hypertension nor associated medical comorbidities. The research protocol was approved by the University of Toronto Research Ethics Board and adhered to the tenets of the Declaration of Helsinki.

### 2.2. Study procedure and data collection

All patients underwent a comprehensive neuro-ophthalmologic examination by a fellowship-trained neuro-ophthalmologist (A.H., E.M., L.D.) that included a thorough history, VA measurement, and dilated fundus examination. A Cirrus spectral-domain OCT unit (Zeiss Cirrus 600 fundus photo/OCT unit) was used to measure the average RNFL thicknesses and to obtain color and autofluorescence fundus photographs. Enhanced depth imaging (EDI-OCT) was performed using the same Cirrus OCT unit with 5 horizontal sections 6 mm in length centered on the optic nerve, spaced 0.25 mm apart vertically in all patients to assess for the presence of buried ODD. Humphrey VF testing (24-2 algorithm) was performed using the Swedish Interactive Thresholding Algorithm Fast protocol. The appearance of the optic nerve head on

ophthalmoscopy was graded using the Frisen scale by one of the raters (A.H.) post-hoc. Magnetic resonance imaging and venography (MRI/MRV) was performed for all patients with specific instructions to the interpreting neuro-radiologist to look for indirect signs of increased ICP (e.g. transverse venous sinus stenosis) and secondary causes of intracranial hypertension (e.g. cerebral venous sinus thrombosis).

### 2.3. Assessment of optic nerves

Fundus photographs were taken at the patient's first presentation and rated at least 6 months after each patient's most recent follow-up to avoid potential recollection of optic nerve appearance (Supplementary File 1). Neuro-ophthalmologist raters were first blinded to all clinical information and presented only with the appearance of the optic nerves on color photographs, and were asked to determine whether the disc swelling represented papilledema or pseudopapilledema. After making this decision, raters were provided with the patient's age, sex, BMI, and VA, and again asked to determine whether the disc swelling represented papilledema or pseudopapilledema. We chose to provide these demographic and clinical factors as any health care provider could obtain these measurements and supply them in a neuro-ophthalmology referral.

Raters' responses were compared with a presumptive diagnosis of papilledema or pseudopapilledema. As LP was not performed for most patients, we used the following strict criteria to define the papilledema cohort: 1) typical demographic for IIH (female, age  $< 50$  years, BMI  $> 30 \text{ kg/m}^2$ ); 2) bilateral TVSS on MRV or  $> 1$  indirect neuroimaging signs of raised ICP on MRI/MRV, and 3) absence of ODD on autofluorescence imaging and EDI-OCT or improvement in symptoms and optic disc swelling following weight loss or ICP-lowering therapy. The pseudopapilledema cohort was defined by 1) visible ODD on autofluorescence imaging or EDI-OCT; 2) patients had normal visual function and no new or worsening symptoms of IIH; and 3) MRI/MRV showed no indirect signs of raised ICP or the optic nerve appearance was stable after at least 1 year of follow-up. All patients underwent MRI/MRV to assess for the presence of indirect neuroimaging signs of intracranial hypertension even if ODD were visible on funduscopy, OCT, and/or autofluorescence imaging, to rule out the possibility of papilledema concurrent with visible or buried ODD. Raters were masked to the overall number of patients with presumed papilledema and pseudopapilledema.

Raters 1, 2, and 3 had a total of 4, 16, and 4 years of neuro-ophthalmic practice at the time of data extraction, respectively. All raters used the same standardized criteria to differentiate papilledema and pseudopapilledema. Raters considered the following features supportive of papilledema: obscuration of vessels, presence of peripapillary heme or fluid, greater degree of optic disc elevation with specific attention to blurring of the nasal disc margin, and venous tortuosity. The following features were supportive of pseudopapilledema: tilted or small/cupless nerve, presence of visible optic disc drusen, irregular or lumpy disc borders, and abnormal vascular branching patterns. An atypical demographic and normal BMI suggested pseudopapilledema in equivocal cases.

### 2.4. Data analysis

Continuous data are reported as means with ranges and standard deviations (SD). To investigate for correlations that may unknowingly influence neuro-ophthalmologists' judgements, pairwise comparisons were performed using Welch's ANOVA for continuous data and the Chi-square test for nominal data. Cohen's and Fleiss's kappa were calculated to assess inter-rater agreement between multiple raters. A negative value was considered no agreement, 0.01–0.20 as slight, 0.21–0.40 as fair, 0.41–0.60 as moderate, 0.61–0.80 as substantial, and 0.81–1.00 as strong agreement. Data analysis was performed using IBM SPSS Statistics Premium 29 (Version 29.0.1.0).

### 3. Results

#### 3.1. Patient population

Fifty-five patients fulfilled the inclusion criteria, consisting of 29 in the papilledema group (52.7 %) and 26 in the pseudopapilledema group (47.3 %). While not the primary aim of the study, three patients underwent LP, and all demonstrated opening pressures consistent with our presumptive diagnosis of papilledema ( $n = 2$ , opening pressures 38 and 27 cm H<sub>2</sub>O) and pseudopapilledema ( $n = 1$ , opening pressure 18 cm H<sub>2</sub>O). Mean age at presentation was 32.8 years (range 18–51, SD 9.3) and 93.7 % ( $n = 51$ ) were female. Average BMI was 30.9 kg/m<sup>2</sup> (range 17.9–51, SD 7.4). Patients with papilledema tended to be younger (29.5 vs 35.8 years,  $p = 0.01$ ) and have lower BMI (28.6 vs 33.0 kg/m<sup>2</sup>,  $p = 0.03$ ) than patients with pseudopapilledema. Two patients in the papilledema group belonged to an atypical demographic for IIH. There were no other statistically significant differences in demographic and clinical characteristics between patients with papilledema and pseudopapilledema (Table 1).

#### 3.2. Accuracy of decisions without clinical information

Inter-rater reliability was 0.052 amongst all raters ( $p = 0.51$ ), 0.13 amongst raters 1 and 2 ( $p = 0.33$ ), 0.055 amongst raters 2 and 3 ( $p = 0.68$ ), and  $-0.25$  amongst raters 1 and 3 ( $p = 0.85$ ). Averaged across all graders, the rate of true positives (i.e. fundus photo of papilledema

correctly identified as papilledema) was 56.3 % and the rate of false negatives (i.e. fundus photo of papilledema incorrectly identified as pseudopapilledema) was 43.7 %. The rate of false positives (i.e. fundus photo of pseudopapilledema incorrectly identified as true papilledema) was 37.2 % and the rate of true negatives (i.e. fundus photo of pseudopapilledema correctly identified as pseudopapilledema) was 62.8 %. Rater 1's, 2's, and 3's decisions agreed with the presumed diagnosis of papilledema/pseudopapilledema in 54.5 %, 65.5 %, and 58.2 % of cases, respectively, corresponding to an average accuracy of 59.4 % (Table 1).

#### 3.3. Accuracy of decisions with the provision of clinical information

After providing the age, sex, BMI, and VA, inter-rater reliability improved to 0.37 amongst all raters ( $p < 0.001$ ), 0.27 amongst raters 1 and 2 ( $p = 0.026$ ), 0.34 amongst raters 2 and 3 ( $p = 0.011$ ), and 0.54 amongst raters 1 and 3 ( $p < 0.001$ ). Raters changed their initial decision in 32.7 %, 16.4 %, and 12.7 % of cases, respectively. Average true positive rate increased to 60.9 %, false negative rate decreased to 39.1 %, false positive rate decreased to 32.1 %, and true negative rate increased to 67.9 %. The changes in overall accuracy for raters 1, 2, and 3 were + 11.0 % (final 65.5 %), +5.4 % (final 70.9 %), and  $-1.8$  % (final 56.4 %), respectively, corresponding to an average improvement of +4.8 % to 64.2 %.

Rater 1 was more likely to rate a patient with higher Frisen grading as papilledema (3.0 vs 1.5,  $p < 0.001$ ); however, the difference lost significance after the provision of clinical information (2.7 vs 1.9,  $p =$

**Table 1**

Demographic, clinical, symptom, and radiologic features of patients with mild papilledema and pseudopapilledema, grouped by rater accuracy before and after being provided with clinical information.

|                          |                                           | Before clinical information     |                                 |                                 | After clinical information      |                                 |                                 |
|--------------------------|-------------------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
|                          | All patients (true/<br>pseudopapilledema) | Rater 1 (correct/<br>incorrect) | Rater 2 (correct/<br>incorrect) | Rater 3 (correct/<br>incorrect) | Rater 1 (correct/<br>incorrect) | Rater 2 (correct/<br>incorrect) | Rater 3 (correct/<br>incorrect) |
|                          | Demographic data                          |                                 |                                 |                                 |                                 |                                 |                                 |
| Age                      | 32.8 (35.8/29.5)                          | 31.9/33.9                       | 32.4/33.6                       | 33.4/32.0                       | 31.9/34.5                       | 33.6/30.9                       | 33.2/32.3                       |
| BMI (kg/m <sup>2</sup> ) | 30.9 (33.0/28.6)                          | 31.2/30.6                       | 29.9/33.0                       | 30.7/31.3                       | 31.1/30.6                       | 29.9/33.4                       | 30.2/31.9                       |
|                          | Clinical data                             |                                 |                                 |                                 |                                 |                                 |                                 |
| VA OD                    | 0.020 (0.024/0.015)                       | 0.020/0.020                     | 0.025/0.011                     | 0.019/0.022                     | 0.017/0.026                     | 0.023/0.013                     | 0.019/0.021                     |
| VA OS                    | 0.025 (0.027/0.023)                       | 0.020/0.031                     | 0.030/0.016                     | 0.022/0.030                     | 0.022/0.031                     | 0.028/0.019                     | 0.026/0.024                     |
| VA OU                    | 0.029 (0.026/0.019)                       | 0.020/0.026                     | 0.028/0.013                     | 0.020/0.026                     | 0.019/0.028                     | 0.025/0.016                     | 0.023/0.023                     |
| VF MD OD (dB)            | −1.87<br>(−2.03/−1.70)                    | −2.08/−1.62                     | −1.96/−1.70                     | −1.72/−2.08                     | −2.01/−1.61                     | −1.81/−2.02                     | −1.87/−1.87                     |
| VF MD OS (dB)            | −2.09<br>(−2.06/−2.13)                    | −2.33/−1.81                     | −2.25/−1.79                     | −1.89/−2.37                     | −2.23/−1.83                     | −2.09/−2.09                     | −2.04/−2.16                     |
| VF MD OU (dB)            | −1.98<br>(−2.04/−1.91)                    | −2.20/−1.71                     | −2.10/−1.75                     | −1.81/−2.23                     | −2.12/−1.72                     | −1.95/−2.05                     | −1.95/−2.01                     |
| RNFL OD (μm)             | 110.3 (113.6/106.7)                       | 108.6/112.3                     | 114.9/101.6                     | 110.8/109.7                     | 111.6/107.7                     | 112.8/104.1                     | 112.6/107.3                     |
| RNFL OS (μm)             | 110.4 (113.4/107.0)                       | 108.1/113.2                     | 114.2/103.2                     | 112.3/107.8                     | 110.9/109.4                     | 113.2/103.8                     | 112.2/108.1                     |
| RNFL OU (μm)             | 110.4 (113.5/106.8)                       | 108.4/112.8                     | 114.5/102.4                     | 111.5/108.7                     | 111.3/108.6                     | 113.0/103.9                     | 112.4/107.7                     |
| Frisen grade             | 2.16 (2.31/2.00)                          | 2.13/2.20                       | 2.22/2.05                       | 2.09/2.26                       | 2.17/2.16                       | 2.10/2.31                       | 2.29/2.00                       |
|                          | Symptomatology                            |                                 |                                 |                                 |                                 |                                 |                                 |
| Headache                 | 23 (11/12)                                | 14/9                            | 16/7                            | 16/7                            | 18/5                            | 17/6                            | 13/10                           |
| Diplopia                 | 0 (0/0)                                   | 0/0                             | 0/0                             | 0/0                             | 0/0                             | 0/0                             | 0/0                             |
| Pulsatile Tinnitus       | 8 (3/5)                                   | 6/2                             | 7/1                             | 6/2                             | 7/1                             | 7/1                             | 6/2                             |
| TVO                      | 3 (1/2)                                   | 1/2                             | 2/1                             | 1/2                             | 1/2                             | 1/2                             | 1/2                             |
|                          | Neuroradiologic data                      |                                 |                                 |                                 |                                 |                                 |                                 |
| TVSS                     | 24 (21/3)                                 | 11/13                           | 16/8                            | 12/12                           | 11/13                           | 20/4                            | 14/10                           |
| Empty sella              | 19 (17/2)                                 | 13/6                            | 16/3                            | 12/7                            | 13/6                            | 16/3                            | 13/6                            |
| Flattened globe          | 12 (10/2)                                 | 5/7                             | 8/4                             | 7/5                             | 8/4                             | 9/3                             | 9/3                             |
| Nerve sheath distension  | 7 (6/1)                                   | 4/3                             | 6/1                             | 4/3                             | 4/3                             | 5/2                             | 5/2                             |
| Nerve tortuosity         | 6 (5/1)                                   | 2/4                             | 5/1                             | 3/3                             | 4/2                             | 5/1                             | 5/1                             |

VA = visual acuity, OD = right eye, OS = left eye, OU = averaged between both eyes, VF MD = visual field mean deviation, RNFL = retinal nerve fiber layer thickness, TVO = transient visual obscuration, TVSS = transverse venous sinus stenosis.

0.08). There was no other demographic, clinical, or radiologic variables that were associated with raters' decisions (Table 2).

#### 4. Discussion

In this cross-sectional analysis of mildly swollen optic discs, neuro-ophthalmologists had poor accuracy (59.4 %) and even poorer inter-rater reliability ( $k = 0.052$ ) when differentiating between papilledema and pseudopapilledema using fundus photographs alone. By providing additional demographic and clinical information, accuracy (64.2 %) and inter-rater agreement ( $k = 0.37$ ) improved.

In tele-ophthalmology, remote evaluation of fundus photographs is becoming commonplace, but the quality and comprehensiveness of referrals is variable [15]. A neuro-ophthalmic referral may be made by other specialties prior to subjecting patients to invasive and potentially harmful diagnostic testing such as LP. A recent study evaluated the ability of physicians from different disciplines to assess fundus photographs of papilledema, pseudopapilledema, and other optic nerve diseases, finding that neuro-ophthalmologists were more accurate than family physicians, neurologists, neurosurgeons, and general ophthalmologists in identifying papilledema and pseudopapilledema. Neuro-ophthalmologists had an average accuracy of 83.5 %, but only 13.0 % of cases had mild disc swelling [14]. The present study demonstrated that neuro-ophthalmologists have much poorer accuracy and inter-rater reliability when optic disc swelling is mild, but that discriminative ability could be improved with the provision of basic patient

information such as age, sex, BMI, and VA. These findings highlight the importance of incorporating detailed demographic and clinical information in neuro-ophthalmology referrals to inform pre-test probability and guide clinical decision-making.

OCT may provide several clues that help to differentiate papilledema from pseudopapilledema. RNFL thickness tends to be greater in cases of true optic disc edema, but in mild disc swelling it is non-specific and often similar between papilledema (Frisen grade 2 or less) and buried ODD [16,17]. Additionally, visualizing an anteriorly-positioned opening angle of Bruch's membrane on SD-OCT may suggest true papilledema, as in pseudopapilledema Bruch's membrane remains flat and follows the natural contours of the eye [18]. EDI-OCT is useful to confirm pseudopapilledema as it may visualize ODD up to 500  $\mu\text{m}$  deeper than what is captured on conventional SD-OCT. [19] EDI-OCT has been shown to be up to 92 % sensitive in detecting buried ODD and may be considered the new gold standard in its diagnosis [20]. In a patient with elevated optic discs but lack of drusen on fundus autofluorescence and EDI-OCT, true disc edema is much more likely. Unfortunately, SD-OCT and EDI-OCT are not readily accessible for optometrists and non-eye care providers who seek neuro-ophthalmic consultation.

In cases of mild optic disc swelling, unless the clinical history and ancillary testing is highly suggestive of pseudopapilledema due to visible ODD, vitreopapillary traction, tilted discs, or other entities, neuroimaging should be performed to exclude secondary causes of intracranial hypertension and identify indirect signs of intracranial hypertension. Bilateral TVSS has been shown to be 84.4 % sensitive and

**Table 2**

Demographic, clinical, symptom, and radiologic features of patients with mild papilledema and pseudopapilledema, grouped by raters' decisions.

|                           | A                      | B                      | C                      | D                      | E                      | F                      |
|---------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| # of patients             | 24/31                  | 18/37                  | 28/27                  | 31/24                  | 26/29                  | 29/26                  |
| Demographic data          |                        |                        |                        |                        |                        |                        |
| Age                       | 33.5/32.3; $p = 1.0$   | 32.2/33.1; $p = 1.0$   | 35.5/30.0; $p = 0.8$   | 35.1/29.9; $p = 0.9$   | 31.9/33.6; $p = 1.0$   | 34.9/30.5; $p = 1.0$   |
| Females                   | 21/30; $p = 1.0$       | 17/34; $p = 1.0$       | 26/25; $p = 1.0$       | 29/22; $p = 1.0$       | 25/26; $p = 1.0$       | 27/24; $p = 1.0$       |
| BMI (kg/m <sup>2</sup> )  | 30.1/31.6; $p = 1.0$   | 35.2/28.8; $p = 0.067$ | 29.7/32.2; $p = 1.0$   | 32.3/29.1; $p = 1.0$   | 33.8/28.4; $p = 0.4$   | 32.6/29.1; $p = 1.0$   |
| Clinical data             |                        |                        |                        |                        |                        |                        |
| VA OD                     | 0.017/0.017; $p = 1.0$ | 0.033/0.014; $p = 1.0$ | 0.025/0.015; $p = 1.0$ | 0.029/0.008; $p = 0.9$ | 0.023/0.017; $p = 1.0$ | 0.021/0.019; $p = 1.0$ |
| VA OS                     | 0.025/0.025; $p = 1.0$ | 0.044/0.016; $p = 0.9$ | 0.024/0.026; $p = 1.0$ | 0.028/0.021; $p = 1.0$ | 0.035/0.017; $p = 1.0$ | 0.035/0.015; $p = 1.0$ |
| VA OU                     | 0.021/0.024; $p = 1.0$ | 0.039/0.015; $p = 0.9$ | 0.025/0.020; $p = 1.0$ | 0.029/0.015; $p = 1.0$ | 0.029/0.017; $p = 1.0$ | 0.028/0.017; $p = 1.0$ |
| VF MD OD (dB)             | -2.13/-1.67; $p = 1.0$ | -2.32/-1.65; $p = 1.0$ | -1.93/-1.81; $p = 1.0$ | -1.72/-2.07; $p = 1.0$ | -1.80/-1.93; $p = 1.0$ | -1.79/-1.96; $p = 1.0$ |
| VF MD OS (dB)             | -2.10/-2.09; $p = 1.0$ | -2.28/-2.00; $p = 1.0$ | -1.96/-2.22; $p = 1.0$ | -1.95/-2.28; $p = 1.0$ | -2.22/-1.97; $p = 1.0$ | -2.11/-2.08; $p = 1.0$ |
| VF MD OU (dB)             | -2.12/-1.88; $p = 1.0$ | -2.30/-1.82; $p = 1.0$ | -1.94/-2.02; $p = 1.0$ | -1.83/-2.17; $p = 1.0$ | -2.01/-1.95; $p = 1.0$ | -1.95/-2.02; $p = 1.0$ |
| RNFL OD ( $\mu\text{m}$ ) | 116.1/105.8; $p = 0.5$ | 115.9/107.6; $p = 0.9$ | 115.6/104.8; $p = 0.5$ | 113.7/105.8; $p = 0.9$ | 112.3/108.4; $p = 1.0$ | 110.5/110.0; $p = 1.0$ |
| RNFL OS ( $\mu\text{m}$ ) | 114.5/107.3; $p = 1.0$ | 115.4/108.0; $p = 1.0$ | 117.0/103.6; $p = 0.4$ | 114.6/105.0; $p = 0.9$ | 114.8/106.4; $p = 1.0$ | 111.6/109.2; $p = 1.0$ |
| RNFL OU ( $\mu\text{m}$ ) | 115.3/106.5; $p = 0.8$ | 115.6/107.8; $p = 0.9$ | 116.3/104.2; $p = 0.8$ | 114.2/105.4; $p = 0.9$ | 113.6/107.4; $p = 1.0$ | 111.0/109.6; $p = 1.0$ |
| Frisen grade              | 3.0/1.5; $p < 0.001$   | 2.7/1.9; $p = 0.2$     | 2.4/1.9; $p = 0.8$     | 2.3/2.0; $p = 1.0$     | 2.2/2.1; $p = 1.0$     | 2.1/2.2; $p = 1.0$     |
| Symptomatology            |                        |                        |                        |                        |                        |                        |
| Headache                  | 8/15                   | 10/13                  | 12/11                  | 11/12                  | 12/11                  | 15/8                   |
| Diplopia                  | 0/0                    | 0/0                    | 0/0                    | 0/0                    | 0/0                    | 0/0                    |
| Pulsatile Tinnitus        | 3/5                    | 4/4                    | 2/6                    | 2/6                    | 3/5                    | 5/3                    |
| TVO                       | 3/0                    | 3/0                    | 2/1                    | 3/0                    | 3/0                    | 3/0                    |
| Neuroradiologic data      |                        |                        |                        |                        |                        |                        |
| TVSS                      | 10/14                  | 8/16                   | 15/9                   | 19/5                   | 13/11                  | 13/11                  |
| Empty sella               | 11/8                   | 11/8                   | 16/3                   | 14/5                   | 10/9                   | 11/8                   |
| Flattened globe           | 7/5                    | 6/6                    | 6/6                    | 7/5                    | 7/5                    | 7/5                    |
| Nerve sheath distension   | 5/2                    | 3/4                    | 5/2                    | 4/3                    | 5/2                    | 4/3                    |
| Nerve tortuosity          | 3/3                    | 3/3                    | 4/2                    | 4/2                    | 4/2                    | 4/2                    |

A = Rater 1 without clinical information (rated as true/pseudopapilledema).

B = Rater 1 with clinical information (rated as true/pseudopapilledema).

C = Rater 2 without clinical information (rated as true/pseudopapilledema).

D = Rater 2 with clinical information (rated as true/pseudopapilledema).

E = Rater 3 without clinical information (rated as true/pseudopapilledema).

F = Rater 3 with clinical information (rated as true/pseudopapilledema).

$p = 1.0$  for all symptomatology and neuroradiologic data.

VA = visual acuity, OD = right eye, OS = left eye, OU = averaged between both eyes, VF MD = visual field mean deviation, RNFL = retinal nerve fiber layer thickness, TVO = transient visual obscuration, TVSS = transverse venous sinus stenosis.



94.9 % specific for increased ICP [21–23]. However, sensitivity and specificity increases when more indirect imaging signs of raised ICP are present [24,25]. For patients who cannot undergo LP for reasons such as patient refusal or lack of access, it is reasonable for an experienced neuro-ophthalmologist to presume a diagnosis of IIH if they belong to the typical demographic for IIH, are systemically well with no red flag symptoms, have normal visual function, and have a normal MRI/MRV with the exception of indirect neuroimaging signs of ICP [8,9,24]. These patients can be treated presumptively and followed in-person at a neuro-ophthalmology clinic. However, atypical features such as optic disc swelling out of proportion to the amount of optic disc drusen require an LP to confirm the diagnosis, which should be ordered as part of the initial telehealth consult.

This study had some limitations. Most patients did not receive an LP as this invasive procedure was rarely warranted. We applied strict criteria to our definitions of papilledema and pseudopapilledema to minimize false positives and false negatives given the lack of an LP. Of note, all 3 patients who ultimately underwent LP matched our presumed diagnosis; however, this number cannot be confidently extrapolated to suggest all patients, had they undergone LP, would have matched our presumed diagnosis. Another limitation is that our definitions of papilledema and pseudopapilledema were absolute and did not account for patients who concurrently have both true disc edema and pseudo-disc swelling. Finally, we had a small number of raters, which prevented inferential statistical analysis of changes in rater accuracy after provision of demographic and clinical information. Future studies with more raters may be conducted to draw stronger conclusions regarding the impact of a detailed referral on the ability to distinguish papilledema from pseudopapilledema.

This cross-sectional study demonstrated that it is difficult to distinguish between papilledema and pseudopapilledema in cases of mild optic disc swelling, even for experienced neuro-ophthalmologists. However, accuracy and inter-rater reliability can be improved with the inclusion of basic clinical and demographic information, highlighting the importance of clinical context in tele-ophthalmology referrals. Patients with mild disc edema who lack evidence of pseudopapilledema on ancillary testing require an MRI/MRV, and if neuroimaging fails to demonstrate indirect signs of raised ICP, then an LP is warranted to confirm the diagnosis.

#### CRedit authorship contribution statement

**Marko Chi-Wei Tien:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jim Xie:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Armin Handzic:** Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Data curation. **Laura Donaldson:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Edward Margolin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

#### Funding

The authors declare no conflicts of interest. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

#### Declaration of competing interest

None.

#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jns.2025.123482>.

#### Data availability

Data and research material will be made available to other researchers from the corresponding author upon reasonable request.

#### References

- [1] B.B. Bruce, C. Lamirel, V. Biousse, et al., Feasibility of nonmydriatic ocular fundus photography in the emergency department: phase I of the FOTO-ED study, *Acad. Emerg. Med.* 18 (9) (2011) 928–933, <https://doi.org/10.1111/j.1553-2712.2011.01147.x>.
- [2] B.B. Bruce, P. Thulasi, C.L. Fraser, et al., Diagnostic accuracy and use of nonmydriatic ocular fundus photography by emergency physicians: phase II of the FOTO-ED study, *Ann. Emerg. Med.* 62 (1) (2013) 28–33.e1, <https://doi.org/10.1016/j.annemergmed.2013.01.010>.
- [3] Y.S. Han, M. Pathipati, C. Pan, et al., Comparison of telemedicine screening of diabetic retinopathy by Mydriatic smartphone-based vs Nonmydriatic tabletop camera-based fundus imaging, *J. Vitreoretin. Dis.* 5 (3) (2020) 199–207, <https://doi.org/10.1177/2474126420958304>.
- [4] E.A. Muro-Fuentes, L. Stunkel, Diagnostic error in neuro-ophthalmology: avenues to improve, *Curr. Neurol. Neurosci. Rep.* 22 (4) (2022) 243–256, <https://doi.org/10.1007/s11910-022-01189-4>.
- [5] M. Wall, D. George, Idiopathic intracranial hypertension. A prospective study of 50 patients, *Brain* 114 (Pt 1A) (1991) 155–180.
- [6] J. Best, G. Silvestri, B. Burton, B. Foot, J. Acheson, The incidence of blindness due to Idiopathic intracranial hypertension in the UK, *Open Ophthalmol. J.* 7 (2013) 26–29, <https://doi.org/10.2174/1874364101307010026>.
- [7] P. Freund, E. Margolin, Pseudopapilledema, in: StatPearls, StatPearls Publishing, 2023. Accessed October 9, 2023, <http://www.ncbi.nlm.nih.gov/books/NBK538291/>.
- [8] H.E. Moss, E.A. Margolin, A.G. Lee, G.P. Van Stavern, Should lumbar puncture be required to diagnose every patient with Idiopathic intracranial hypertension? *J. Neuroophthalmol.* 41 (3) (2021) 379–384, <https://doi.org/10.1097/WNO.0000000000001373>.
- [9] A.R. Vosoughi, E.A. Margolin, J.A. Miceli, Can lumbar puncture be safely deferred in patients with mild presumed Idiopathic intracranial hypertension? *J. Neuroophthalmol.* 42 (4) (2022) 505–508, <https://doi.org/10.1097/WNO.0000000000001411>.
- [10] M.S. Margolis, A.A. DeBusk, M.L. Moster, et al., Lumbar puncture for diagnosis of Idiopathic intracranial hypertension in typical patients, *J. Neuroophthalmol.* 41 (3) (2021) 375–378, <https://doi.org/10.1097/WNO.0000000000001319>.
- [11] C.R. Durst, D.A. Ornan, M.A. Reardon, et al., Prevalence of dural venous sinus stenosis and hypoplasia in a generalized population, *J. Neurointerv. Surg.* 8 (11) (2016) 1173–1177, <https://doi.org/10.1136/neurintsurg-2015-012147>.
- [12] C.R. Durst, A.M. Saindane, J.H. Peragallo, B.B. Bruce, N.J. Newman, V. Biousse, Brain imaging in Idiopathic intracranial hypertension, *J. Neuroophthalmol.* 35 (4) (2015) 400–411, <https://doi.org/10.1097/WNO.0000000000000303>.
- [13] B.S. Chen, B.I. Meyer, A.M. Saindane, B.B. Bruce, N.J. Newman, V. Biousse, Prevalence of incidentally detected signs of intracranial hypertension on magnetic resonance imaging and their association with papilledema, *JAMA Neurol.* 78 (6) (2021) 718–725, <https://doi.org/10.1001/jamaneurol.2021.0710>.
- [14] L.N. Johnson, R.S. Hepler, M.J. Bartholomew, Accuracy of papilledema and pseudopapilledema detection: a multispecialty study, *J. Fam. Pract.* 33 (4) (1991) 381–386.
- [15] V. Khou, A. Ly, L. Moore, et al., Review of referrals reveal the impact of referral content on the triage and management of ophthalmology wait lists, *BMJ Open* 11 (9) (2021) e047246, <https://doi.org/10.1136/bmjopen-2020-047246>.
- [16] K.M. Kulkarni, J. Pasol, P.R. Rosa, B.L. Lam, Differentiating mild papilledema and buried optic nerve head drusen using spectral domain optical coherence tomography, *Ophthalmology* 121 (4) (2014) 959–963, <https://doi.org/10.1016/j.ophtha.2013.10.036>.
- [17] C.V. Vartin, A.M. Nguyen, T. Balmittgere, M. Bernard, C. Tilikete, A. Vighetto, Detection of mild papilloedema using spectral domain optical coherence tomography, *Br. J. Ophthalmol.* 96 (3) (2012) 375–379, <https://doi.org/10.1136/bjo.2010.199562>.
- [18] M.J. Kupersmith, P. Sibony, G. Mandel, M. Durbin, R.H. Kardon, Optical coherence tomography of the swollen optic nerve head: deformation of the Peripapillary retinal pigment epithelium layer in papilledema, *Invest. Ophthalmol. Vis. Sci.* 52 (9) (2011) 6558, <https://doi.org/10.1167/iovs.10-6782>.
- [19] S. Mrejen, R.F. Spaide, Optical coherence tomography: imaging of the choroid and beyond, *Surv. Ophthalmol.* 58 (5) (2013) 387–429, <https://doi.org/10.1016/j.survophthal.2012.12.001>.
- [20] S. Youn, B. Loshusan, J.J. Armstrong, J.A. Fraser, S. Hamann, L.L.C.D. Bursztyn, A comparison of diagnostic accuracy of imaging modalities to detect optic disc Drusen: the age of enhanced depth imaging optical coherence tomography, *Am. J. Ophthalmol.* 248 (2023) 137–144, <https://doi.org/10.1016/j.ajo.2022.12.004>.

- [21] B.D. Riggeal, B.B. Bruce, A.M. Saindane, et al., Clinical course of idiopathic intracranial hypertension with transverse sinus stenosis, *Neurology* 80 (3) (2013) 289–295, <https://doi.org/10.1212/WNL.0b013e31827debd6>.
- [22] R.M. Kwee, T.C. Kwee, Systematic review and meta-analysis of MRI signs for diagnosis of idiopathic intracranial hypertension, *Eur. J. Radiol.* 116 (2019) 106–115, <https://doi.org/10.1016/j.ejrad.2019.04.023>.
- [23] P.P. Morris, D.F. Black, J. Port, N. Campeau, Transverse sinus stenosis is the Most sensitive MR imaging correlate of Idiopathic intracranial hypertension, *AJNR Am. J. Neuroradiol.* 38 (3) (2017) 471–477, <https://doi.org/10.3174/ajnr.A5055>.
- [24] M.T.M. Wang, Z.J. Prime, W. Xu, et al., Diagnostic performance of neuroimaging in suspected idiopathic intracranial hypertension, *J. Clin. Neurosci.* 96 (2022) 56–60, <https://doi.org/10.1016/j.jocn.2021.12.024>.
- [25] N. Prabhat, S. Chandel, D.A. Takkar, et al., Sensitivity and specificity of neuroimaging signs in patients with idiopathic intracranial hypertension, *Neuroradiol. J.* 34 (5) (2021) 421–427, <https://doi.org/10.1177/19714009211000623>.